ROY SUMURUNG SIGALINGGING

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Profile

Information Technology graduate from Del Institute of Technology with a strong interest in Full Stack Development and professional experience as a Full Stack Developer at PT Citra Bonang, contributing to a web application migration from version 1.0 to 2.0 through feature development, system enhancement, and frontend-backend integration. Skilled in developing web applications using Laravel, React.js, and Next.js, building mobile applications with Flutter, managing MySQL databases, and integrating REST APIs, with additional experience in Machine Learning implementation. with strong problem-solving and teamwork skills, eager to learn new technologies and contribute to developing innovative and effective digital solutions.

Education

Del Institute of Technology September 2023 – September 2026
Associate Degree in Information Technology
GPA : 3.26 / 4.00

Work Experience

PT Citra Bonang February 2026 – July 2026
Full Stack Developer
Tech Stack: *CodeIgniter, PHP, jQuery EasyUI, JavaScript, MySQL*

- Migrated the web application from Version 1.0 to Version 2.0.
- Developed and customized web application features to meet the company's requirements.
- Improved overall system functionality and performance.
- Ensured seamless integration between the frontend, backend, and database.

Project Experiences

Design and Development of the Rumah Damai Children's Foundation Information Website January 2024 – June 2024
Full Stack Developer
Tech Stack: *Laravel, PHP, Blade Templates, HTML, CSS, JavaScript, MySQL*

- Developed a responsive information website for Rumah Damai Children’s Foundation with Laravel.
- Built the front end with Blade Templates, HTML, CSS, and JavaScript to present foundation information clearly to visitors.
- Implemented backend functionality including application logic, routing, controllers, and database integration.
- Handled the full development cycle from design to deployment

Design & Development of the Pusmasib Mobile Application January 2024 – June 2024

Full Stack Developer

Tech Stack: *Flutter, Dart, Laravel, PHP, MySQL, REST API*

- Developed a cross-platform mobile application with Flutter and Dart with smooth navigation and a clear user interface.
- Built the backend with Laravel, implementing business logic, routing, and controllers.
- Designed and managed the MySQL database schema for efficient data storage and retrieval.
- Connected the mobile app to the Laravel backend through REST APIs, enabling reliable data exchange between mobile and web systems.

Application of CNN Algorithm with MobileNetV4 Architecture for Cataract Classification August 2025 – January 2026

Machine Learning Engineer

Tech Stack: *Python, PyTorch, MobileNetV4, CNN, Optuna, Flutter, Dart*

- Built a deep learning model with PyTorch and MobileNetV4 to classify eye images into three categories: Normal, Immature, and Mature cataract.
- Designed an end-to-end machine learning pipeline covering data preparation, training, validation, testing, and performance evaluation.
- Improved model performance through fine-tuning and hyperparameter optimization with Optuna (learning rate, batch size, dropout, weight decay, and classifier configuration).
- Deployed the model in a Flutter mobile application that lets users classify cataract conditions from eye images.

Mobile Application Development Project: Movie Catalog August 2024 – December 2024

Mobile Application Developer

Tech Stack: *Flutter, Dart, REST API, SharedPreferences, SQLite*

- Developed a mobile application using Flutter and Dart, covering the entire development process from UI design to feature implementation and data integration.
- Built responsive, modern, and user-friendly interfaces using Flutter widgets to ensure an optimal user experience across various devices and screen sizes.
- Implemented state management, navigation flows, and application logic to support stable and efficient user interactions.
- Integrated the application with APIs and local storage solutions, such as SharedPreferences and SQLite, to ensure efficient data management.

Skills

- Programming Language: C, C#, PHP, Java, Python, Golang, JavaScript, React.js, TypeScript, Dart.
- Framework & Libraries: Laravel, FastAPI, ReactJs, Flutter, Flask, NodeJs, NextJs, Easyui.
- Testing Tools: Manual Testing, Selenium, Katalon.
- Database Management: MySQL, PostgreSQL, MongoDB.
- Clouds: Google Cloud Service.
- Others: Git, Github, Trello, Figma, Balsamic, Android Studio, Postman, NetBeans.

Organizational Experience

- Facilities and Social Division of HIMATIF — August 2024 – August 2026
- Del Data Science — August 2024 – December 2024

Course Taken & Certificate

- Toefl ITP (467)
- Google AI Essentials
- Share Data Through the Art of Visualization